

Sequences and Series of Real Numbers (2015–2021)

Category: Real Analysis • Official Mock Test Series

TOTAL QUESTIONS

55

TOTAL MARKS

90 M

TEST DURATION

165 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2015–20)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	26	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	8	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	21	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2021 • Q55 • ID: JAM_2021_Q55)

NAT

+2 M

Neg: Nil (0)

Q. 55 Define the sequence

$$s_n = \begin{cases} \frac{1}{2^n} \sum_{j=0}^{n-2} 2^{2j} & \text{if } n > 0 \text{ is even,} \\ \frac{1}{2^n} \sum_{j=0}^{n-1} 2^{2j} & \text{if } n > 0 \text{ is odd.} \end{cases}$$



Define $\sigma_m = \frac{1}{m} \sum_{n=1}^m s_n$. The number of limit points of the sequence $\{\sigma_m\}$ is ____.

MA

16 / 17

Question 2 (JAM 2021 • Q50 • ID: JAM_2021_Q50)

NAT

+1 M

Neg: Nil (0)

Q. 50 The value of

$$\frac{\pi}{2} \lim_{n \rightarrow \infty} \cos\left(\frac{\pi}{4}\right) \cos\left(\frac{\pi}{8}\right) \cdots \cos\left(\frac{\pi}{2^{n+1}}\right)$$

is ____.



MA

15 / 17

Question 3 (JAM 2021 • Q42 • ID: JAM_2021_Q42)

NAT

+1 M

Neg: Nil (0)

Q. 42 The value of

$$\lim_{n \rightarrow \infty} \left(3^n + 5^n + 7^n \right)^{\frac{1}{n}}$$

is ____.

Question 4 (JAM 2021 • Q30 • ID: JAM_2021_Q30)

MCQ

+2 M

Neg: -0.67

Q. 30 Consider the two series

$$\text{I. } \sum_{n=1}^{\infty} \frac{1}{n^{1+(1/n)}} \quad \text{and} \quad \text{II. } \sum_{n=1}^{\infty} \frac{1}{n^{2-n^{1/n}}}$$

Which one of the following holds?

(A) Both I and II converge.

(B) Both I and II diverge.

(C) I converges and II diverges.

(D) I diverges and II converges.



MA

10 / 17

Question 5 (JAM 2021 • Q16 • ID: JAM_2021_Q16)

MCQ

+2 M

Neg: -0.67

Q. 16 Define

$$S = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{2^2}\right) \left(1 - \frac{1}{3^2}\right) \cdots \left(1 - \frac{1}{n^2}\right).$$

Then

(A) $S = 1/2$.(B) $S = 1/4$.(C) $S = 1$.(D) $S = 3/4$.

Question 6 (JAM 2021 • Q15 • ID: JAM_2021_Q15)

MCQ

+2 M

Neg: -0.67

JAM 2021

MATHEMATICS - MA

Q. 15 Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a bijective map such that

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2} < +\infty.$$

The number of such bijective maps is

(A) exactly one.

(B) zero.

(C) finite but more than one.

(D) infinite.

Question 7 (JAM 2020 • Q52 • ID: JAM_2020_Q52)

NAT

+2 M

Neg: Nil (0)

Q. 52 Consider the expansion of the function $f(x) = \frac{3}{(1-x)(1+2x)}$ in powers of x , that is valid in $|x| < \frac{1}{2}$. Then the coefficient of x^4 is _____.

Question 8 (JAM 2020 • Q51 • ID: JAM_2020_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 The sum of the series $\frac{1}{2(2^2-1)} + \frac{1}{3(3^2-1)} + \frac{1}{4(4^2-1)} + \cdots$ is _____.

Question 9 (JAM 2020 • Q41 • ID: JAM_2020_Q41)

NAT

+1 M

Neg: Nil (0)

Q. 41 Let $x_n = n^{\frac{1}{n}}$ and $y_n = e^{1-x_n}$, $n \in \mathbb{N}$. Then the value of $\lim_{n \rightarrow \infty} y_n$ is _____.

Question 10 (JAM 2020 • Q39 • ID: JAM_2020_Q39)

MSQ

+2 M

Neg: Nil (0)

Q. 39 Let f be a real valued function of a real variable, such that $|f^{(n)}(0)| \leq K$ for all $n \in \mathbb{N}$, where $K > 0$. Which of the following is/are true?

- (A) $\left| \frac{f^{(n)}(0)}{n!} \right|^{\frac{1}{n}} \rightarrow 0$ as $n \rightarrow \infty$
- (B) $\left| \frac{f^{(n)}(0)}{n!} \right|^{\frac{1}{n}} \rightarrow \infty$ as $n \rightarrow \infty$
- (C) $f^{(n)}(x)$ exists for all $x \in \mathbb{R}$ and for all $n \in \mathbb{N}$
- (D) The series $\sum_{n=1}^{\infty} \frac{f^{(n)}(0)}{(n-1)!}$ is absolutely convergent

Question 11 (JAM 2020 • Q37 • ID: JAM_2020_Q37)

MSQ

+2 M

Neg: Nil (0)

JAM 2020

MATHEMATICS - MA

Q. 37 Let $a = \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + \cdots + \frac{(n-1)}{n^2} \right)$ and $b = \lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n} \right)$.

Which of the following is/are true?

- (A) $a > b$ (B) $a < b$ (C) $ab = \ln \sqrt{2}$ (D) $\frac{a}{b} = \ln \sqrt{2}$

Question 12 (JAM 2020 • Q32 • ID: JAM_2020_Q32)

MSQ

+2 M

Neg: Nil (0)

Q. 32 If $s_n = \frac{(-1)^n}{2^n + 3}$ and $t_n = \frac{(-1)^n}{4n-1}$, $n = 0, 1, 2, \dots$, then

- (A) $\sum_{n=0}^{\infty} s_n$ is absolutely convergent (B) $\sum_{n=0}^{\infty} t_n$ is absolutely convergent
- (C) $\sum_{n=0}^{\infty} s_n$ is conditionally convergent (D) $\sum_{n=0}^{\infty} t_n$ is conditionally convergent

Question 13 (JAM 2020 • Q9 • ID: JAM_2020_Q9)

MCQ

+1 M

Neg: -0.33

Q. 9 The radius of convergence of the power series

$$\sum_{n=1}^{\infty} \left(\frac{n+2}{n} \right)^{n^2} x^n$$

is

(A) e^2

(B) $\frac{1}{\sqrt{e}}$

(C) $\frac{1}{e}$

(D) $\frac{1}{e^2}$

MA

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Question 14 (JAM 2020 • Q2 • ID: JAM_2020_Q2)

MCQ

+1 M

Neg: -0.33

Q. 2 Let $f(x) = 2x^3 - 9x^2 + 7$. Which of the following is true?(A) f is one-one in the interval $[-1, 1]$ (B) f is one-one in the interval $[2, 4]$ (C) f is NOT one-one in the interval $[-4, 0]$ (D) f is NOT one-one in the interval $[0, 4]$

Question 15 (JAM 2019 • Q52 • ID: JAM_2019_Q52)

NAT

+2 M

Neg: Nil (0)

Q.52 The greatest lower bound of the set

$$\{(e^n + 2^n)^{\frac{1}{n}} : n \in \mathbb{N}\},$$

(round off to 2 decimal places) is _____

Question 16 (JAM 2019 • Q40 • ID: JAM_2019_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40 Let $\{a_n\}$ be the sequence of real numbers such that

$$a_1 = 1 \text{ and } a_{n+1} = a_n + a_n^2 \text{ for all } n \geq 1.$$

Then

(A) $a_4 = a_1(1 + a_1)(1 + a_2)(1 + a_3)$

(B) $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$

(C) $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 1$

(D) $\lim_{n \rightarrow \infty} a_n = 0$

MA

9/13

Question 17 (JAM 2019 • Q36 • ID: JAM_2019_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36 Let $\{a_n\}$ be the sequence given by

$$a_n = \max\left\{\sin\left(\frac{n\pi}{3}\right), \cos\left(\frac{n\pi}{3}\right)\right\}, \quad n \geq 1.$$

Then which of the following statements is/are TRUE about the subsequences $\{a_{6n-1}\}$ and $\{a_{6n+4}\}$?

(A) Both the subsequences are convergent

(B) Only one of the subsequences is convergent

(C) $\{a_{6n-1}\}$ converges to $-\frac{1}{2}$ (D) $\{a_{6n+4}\}$ converges to $\frac{1}{2}$

MA

8/13

Question 18 (JAM 2019 • Q31 • ID: JAM_2019_Q31)

MSQ

+2 M

Neg: Nil (0)

Q. 31 – Q. 40 carry two marks each.Q.31 Let G be a noncyclic group of order 4. Consider the statements I and II:I. There is NO injective (one-one) homomorphism from G to $\mathbb{Z}_8$ II. There is NO surjective (onto) homomorphism from $\mathbb{Z}_8$ to G

Then

(A) I is true

(B) I is false

(C) II is true

(D) II is false

Question 19 (JAM 2019 • Q22 • ID: JAM_2019_Q22)

MCQ

+2 M

Neg: -0.67

Q.22 The function

$$f(x, y) = x^3 + 2xy + y^3$$

has a saddle point at

- (A) $(0, 0)$ (B) $\left(-\frac{2}{3}, -\frac{2}{3}\right)$ (C) $\left(-\frac{3}{2}, -\frac{3}{2}\right)$ (D) $(-1, -1)$

Question 20 (JAM 2019 • Q15 • ID: JAM_2019_Q15)

MCQ

+2 M

Neg: -0.67

Q.15 For $-1 < x < 1$, the sum of the power series $1 + \sum_{n=2}^{\infty} (-1)^{n-1} n^2 x^{n-1}$ is

- (A) $\frac{1-x}{(1+x)^3}$ (B) $\frac{1+x^2}{(1+x)^4}$
 (C) $\frac{1-x}{(1+x)^2}$ (D) $\frac{1+x^2}{(1+x)^3}$

Question 21 (JAM 2019 • Q6 • ID: JAM_2019_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 If $x^h y^k$ is an integrating factor of the differential equation

$$y(1 + xy) dx + x(1 - xy) dy = 0,$$

then the ordered pair (h, k) is equal to

- (A) $(-2, -2)$ (B) $(-2, -1)$ (C) $(-1, -2)$ (D) $(-1, -1)$

Question 22 (JAM 2018 • Q51 • ID: JAM_2018_Q51)

NAT

+2 M

Neg: Nil (0)

Q. 51 – Q. 60 carry two marks each.Q.51 Let $a_k = (-1)^{k-1}$, $s_n = a_1 + a_2 + \cdots + a_n$ and $\sigma_n = (s_1 + s_2 + \cdots + s_n)/n$, where $k, n \in \mathbb{N}$.Then $\lim_{n \rightarrow \infty} \sigma_n$ is _____ (correct up to one decimal place).

Question 23 (JAM 2018 • Q50 • ID: JAM_2018_Q50)**NAT**

+1 M

Neg: Nil (0)

Q.50 The coefficient of x^4 in the power series expansion of $e^{\sin x}$ about $x = 0$ is _____ (correct up to three decimal places).

Q 51 – Q 60 carry two marks each

Question 24 (JAM 2018 • Q47 • ID: JAM_2018_Q47)**NAT**

+1 M

Neg: Nil (0)

Q.47 Let $a_n = \frac{(1+(-1)^n)}{2^n} + \frac{(1+(-1)^{n-1})}{3^n}$. Then the radius of convergence of the power series $\sum_{n=1}^{\infty} a_n x^n$ about $x = 0$ is _____

Question 25 (JAM 2018 • Q43 • ID: JAM_2018_Q43)**NAT**

+1 M

Neg: Nil (0)

Q.43 Let $f(x) = \sum_{n=0}^{\infty} (-1)^n x(x-1)^n$ for $0 < x < 2$. Then the value of $f\left(\frac{\pi}{4}\right)$ is _____

Question 26 (JAM 2018 • Q23 • ID: JAM_2018_Q23)**MCQ**

+2 M

Neg: -0.67

Q.23 An integrating factor of the differential equation $\left(y + \frac{1}{3}y^3 + \frac{1}{2}x^2\right)dx + \frac{1}{4}(x + xy^2)dy = 0$ is

(A) x^2 (B) $3 \log_e x$ (C) x^3 (D) $2 \log_e x$

Question 27 (JAM 2018 • Q13 • ID: JAM_2018_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 Let $a_n = n + \frac{1}{n}$, $n \in \mathbb{N}$. Then the sum of the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{a_{n+1}}{n!}$ is

- (A) $e^{-1} - 1$ (B) e^{-1} (C) $1 - e^{-1}$ (D) $1 + e^{-1}$

Question 28 (JAM 2018 • Q3 • ID: JAM_2018_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 If $\{v_1, v_2, v_3\}$ is a linearly independent set of vectors in a vector space over $\mathbb{R}$, then which one of the following sets is also linearly independent?

- (A) $\{v_1 + v_2 - v_3, 2v_1 + v_2 + 3v_3, 5v_1 + 4v_2\}$
(B) $\{v_1 - v_2, v_2 - v_3, v_3 - v_1\}$
(C) $\{v_1 + v_2 - v_3, v_2 + v_3 - v_1, v_3 + v_1 - v_2, v_1 + v_2 + v_3\}$
(D) $\{v_1 + v_2, v_2 + 2v_3, v_3 + 3v_1\}$

Question 29 (JAM 2017 • Q59 • ID: JAM_2017_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59 The radius of convergence of the power series

$$\sum_{n=0}^{\infty} n! x^{n^2}$$

is _____

Question 30 (JAM 2017 • Q55 • ID: JAM_2017_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55 Let $a_n = \sqrt{n}$, $n \geq 1$, and let $s_n = a_1 + a_2 + \cdots + a_n$. Then

$$\lim_{n \rightarrow \infty} \left(\frac{a_n/s_n}{-\ln(1 - a_n/s_n)} \right) = \underline{\hspace{2cm}}$$

Question 31 (JAM 2017 • Q47 • ID: JAM_2017_Q47)

NAT

+1 M

Neg: Nil (0)

Q.47
$$\frac{1}{2\pi} \left(\frac{\pi^3}{1!3} - \frac{\pi^5}{3!5} + \frac{\pi^7}{5!7} - \cdots + \frac{(-1)^{n-1} \pi^{2n+1}}{(2n-1)!(2n+1)} + \cdots \right) = \underline{\hspace{2cm}}$$

Question 32 (JAM 2017 • Q36 • ID: JAM_2017_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36 Let $\{x_n\}$ be a real sequence such that $7x_{n+1} = x_n^3 + 6$ for $n \geq 1$. Then which of the following statements is/are TRUE?

- (A) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 1
- (B) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 2
- (C) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to 1
- (D) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to -3

Question 33 (JAM 2017 • Q28 • ID: JAM_2017_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 Which one of the following is TRUE?

- (A) Every sequence that has a convergent subsequence is a Cauchy sequence
- (B) Every sequence that has a convergent subsequence is a bounded sequence
- (C) The sequence $\{\sin n\}$ has a convergent subsequence
- (D) The sequence $\left\{n \cos \frac{1}{n}\right\}$ has a convergent subsequence

Question 34 (JAM 2017 • Q27 • ID: JAM_2017_Q27)

MCQ

+2 M

Neg: -0.67

Q.27

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(\frac{1}{\sqrt{3} + \sqrt{6}} + \frac{1}{\sqrt{6} + \sqrt{9}} + \cdots + \frac{1}{\sqrt{3n} + \sqrt{3n+3}} \right) =$$

(A) $1 + \sqrt{3}$

(B) $\sqrt{3}$

(C) $\frac{1}{\sqrt{3}}$

(D) $\frac{1}{1+\sqrt{3}}$

Question 35 (JAM 2017 • Q26 • ID: JAM_2017_Q26)

MCQ

+2 M

Neg: -0.67

Q.26 Let $0 < a_1 < b_1$. For $n \geq 1$, define

$$a_{n+1} = \sqrt{a_n b_n} \quad \text{and} \quad b_{n+1} = \frac{a_n + b_n}{2}.$$

Then which one of the following is NOT TRUE?

(A) Both $\{a_n\}$ and $\{b_n\}$ converge, but the limits are not equal(B) Both $\{a_n\}$ and $\{b_n\}$ converge and the limits are equal(C) $\{b_n\}$ is a decreasing sequence(D) $\{a_n\}$ is an increasing sequence

Question 36 (JAM 2017 • Q21 • ID: JAM_2017_Q21)

MCQ

+2 M

Neg: -0.67

Q.21 The interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{1}{(-3)^{n+2}} \frac{(4x-12)^n}{n^2+1}$$

is

(A) $\frac{10}{4} \leq x < \frac{14}{4}$

(B) $\frac{9}{4} \leq x < \frac{15}{4}$

(C) $\frac{10}{4} \leq x \leq \frac{14}{4}$

(D) $\frac{9}{4} \leq x \leq \frac{15}{4}$

Question 37 (JAM 2017 • Q13 • ID: JAM_2017_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 Let $M = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 2 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Then

$$\lim_{n \rightarrow \infty} M^n x$$

(A) does not exist

(B) is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(C) is $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$

(D) is $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

Question 38 (JAM 2017 • Q9 • ID: JAM_2017_Q9)

MCQ

+1 M

Neg: -0.33

Q.9 If $\lim_{T \rightarrow \infty} \int_0^T e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$, then

$$\lim_{T \rightarrow \infty} \int_0^T x^2 e^{-x^2} dx =$$

(A) $\frac{\sqrt{\pi}}{4}$

(B) $\frac{\sqrt{\pi}}{2}$

(C) $\sqrt{2\pi}$

(D) $2\sqrt{\pi}$

Question 39 (JAM 2017 • Q4 • ID: JAM_2017_Q4)

MCQ

+1 M

Neg: -0.33

Q.4

$$\lim_{n \rightarrow \infty} \frac{\pi}{n} \sum_{k=1}^n \sin\left(\frac{\pi}{2} + \frac{5\pi}{2} \cdot \frac{k}{n}\right) =$$

(A) $\frac{2\pi}{5}$

(B) $\frac{5}{2}$

(C) $\frac{2}{5}$

(D) $\frac{5\pi}{2}$

Question 40 (JAM 2016 • Q51 • ID: JAM_2016_Q51)

NAT

+2 M

Neg: Nil (0)

Q.51 The value of $\lim_{n \rightarrow \infty} \left(8n - \frac{1}{n}\right)^{\frac{(-1)^n}{n^2}}$ is equal to _____

Question 41 (JAM 2016 • Q48 • ID: JAM_2016_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 The radius of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{(-4)^n}{n(n+1)} (x+2)^{2n} \text{ is } \underline{\hspace{2cm}}$$

Question 42 (JAM 2016 • Q42 • ID: JAM_2016_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42 Let $\{s_k\}$ be a sequence of real numbers, where

$$s_k = k^{\alpha/k}, \quad k \geq 1, \quad \alpha > 0.$$

Then

$$\lim_{n \rightarrow \infty} (s_1 s_2 \dots s_n)^{1/n}$$

is _____

MA

10/13

Question 43 (JAM 2016 • Q41 • ID: JAM_2016_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41 Let $\{s_n\}$ be a sequence of real numbers given by

$$s_n = 2^{(-1)^n} \left(1 - \frac{1}{n}\right) \sin \frac{n\pi}{2}, \quad n \in \mathbb{N}.$$

Then the least upper bound of the sequence $\{s_n\}$ is _____

Question 44 (JAM 2016 • Q23 • ID: JAM_2016_Q23)

MCQ

+2 M

Neg: -0.67

Q.23 Let $S \subset \mathbb{R}$ and ∂S denote the set of points x in $\mathbb{R}$ such that every neighbourhood of x contains some points of S as well as some points of complement of S . Further, let $\bar{S}$ denote the closure of S . Then which one of the following is FALSE?

- (A) $\partial \mathbb{Q} = \mathbb{R}$
- (B) $\partial(\mathbb{R} \setminus T) = \partial T, \quad T \subset \mathbb{R}$
- (C) $\partial(T \cup V) = \partial T \cup \partial V, \quad T, V \subset \mathbb{R}, \quad T \cap V \neq \emptyset$
- (D) $\partial T = \bar{T} \cap (\mathbb{R} \setminus T), \quad T \subset \mathbb{R}$

Question 45 (JAM 2016 • Q13 • ID: JAM_2016_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 The largest eigenvalue of the matrix $\begin{bmatrix} 1 & 4 & 16 \\ 4 & 16 & 1 \\ 16 & 1 & 4 \end{bmatrix}$ is

- (A) 16 (B) 21
(C) 48 (D) 64

Question 46 (JAM 2016 • Q6 • ID: JAM_2016_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 The value of $\lim_{(x,y) \rightarrow (2,-2)} \frac{\sqrt{(x-y)}-2}{x-y-4}$ is

- (A) 0 (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Question 47 (JAM 2016 • Q1 • ID: JAM_2016_Q1)

MCQ

+1 M

Neg: -0.33

Q. 1 – Q.10 carry one mark each.

Q.1 The sequence $\{s_n\}$ of real numbers given by

$$s_n = \frac{\sin \frac{\pi}{2}}{1 \cdot 2} + \frac{\sin \frac{\pi}{2^2}}{2 \cdot 3} + \dots + \frac{\sin \frac{\pi}{2^n}}{n \cdot (n+1)}$$

is

- (A) a divergent sequence
(B) an oscillatory sequence
(C) not a Cauchy sequence
(D) a Cauchy sequence

Question 48 (JAM 2015 • Q57 • ID: JAM_2015_Q57)

NAT

+2 M

Neg: Nil (0)

Q.17

If $\int_0^x (e^{-t^2} + \cos t) dt$ has the power series expansion $\sum_{n=1}^{\infty} a_n x^n$, then a_5 , correct upto three decimal places, is equal to _____

Question 49 (JAM 2015 • Q53 • ID: JAM_2015_Q53)

NAT

+2 M

Neg: Nil (0)

Q.13 The limit

is equal to _____

$$\lim_{n \rightarrow \infty} \sum_{k=2}^n \frac{1}{k^3 - k}$$

Question 50 (JAM 2015 • Q46 • ID: JAM_2015_Q46)

NAT

+1 M

Neg: Nil (0)

Q.6

If the power series

$$\sum_{n=0}^{\infty} \frac{n!}{n^n} x^{2n}$$

converges for $|x| < c$ and diverges for $|x| > c$, then the value of c , correct upto three decimal places, is _____

Question 51 (JAM 2015 • Q33 • ID: JAM_2015_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.3

Which of the following conditions implies (imply) the convergence of a sequence $\{x_n\}$ of real numbers?

- (A) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $|x_{n+1} - x_n| < \varepsilon$
- (B) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $\frac{1}{(n+1)^2} |x_{n+1} - x_n| < \varepsilon$
- (C) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, $(n+1)^2 |x_{n+1} - x_n| < \varepsilon$
- (D) Given $\varepsilon > 0$ there exists an $n_0 \in \mathbb{N}$ such that for all m, n with $m > n \geq n_0$, $|x_m - x_n| < \varepsilon$

Question 52 (JAM 2015 • Q23 • ID: JAM_2015_Q23)

MCQ

+2 M

Neg: -0.67

Q.23

The sequence $\left\{ \cos \left(\frac{1}{2} \tan^{-1} \left(-\frac{n}{2} \right)^n \right) \right\}$ is

- (A) monotone and convergent
- (B) monotone but not convergent
- (C) convergent but not monotone
- (D) neither monotone nor convergent

Question 53 (JAM 2015 • Q22 • ID: JAM_2015_Q22)

MCQ

+2 M

Neg: -0.67

Q.22

Which one of the following statements is true for the series $\sum_{n=1}^{\infty} (-1)^n \frac{(2n)!}{n^{2n}}$?

- (A) The series converges conditionally but not absolutely
- (B) The series converges absolutely
- (C) The sequence of partial sums of the series is bounded but not convergent
- (D) The sequence of partial sums of the series is unbounded

Question 54 (JAM 2015 • Q13 • ID: JAM_2015_Q13)

MCQ

+2 M

Neg: -0.67

Q.13 Let G be a nonabelian group. Let $\alpha \in G$ have order 4 and let $\beta \in G$ have order 3. Then the order of the element $\alpha\beta$ in G

- (A) is 6
- (B) is 12
- (C) is of the form $12k$ for $k \geq 2$
- (D) need not be finite

MA

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Question 55 (JAM 2015 • Q6 • ID: JAM_2015_Q6)

MCQ

+1 M

Neg: -0.33

Q.6 The volume of the portion of the solid cylinder $x^2 + y^2 \leq 2$ bounded above by the surface $z = x^2 + y^2$ and bounded below by the xy -plane is

- (A) π
- (B) 2π
- (C) 3π
- (D) 4π

MA

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OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Sequences and Series of Real Numbers (2015–2021) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	0
2	NAT	+1	1
3	NAT	+1	7
4	MCQ	+2	B
5	MCQ	+2	A
6	MCQ	+2	B
7	NAT	+2	33 to 33
8	NAT	+2	0.25 to 0.25
9	NAT	+1	1 to 1
10	MSQ	+2	A;D
11	MSQ	+2	B;C
12	MSQ	+2	A;D
13	MCQ	+1	D
14	MCQ	+1	D
15	NAT	+2	2.69 to 2.74
16	MSQ	+2	A, B
17	MSQ	+2	A
18	MSQ	+2	A, C
19	MCQ	+2	A

Q. No.	Type	Marks	Official Key
20	MCQ	+2	A
21	MCQ	+1	A
22	NAT	+2	0.4 to 0.6
23	NAT	+1	−0.130 to −0.120
24	NAT	+1	2 to 2
25	NAT	+1	1 to 1
26	MCQ	+2	C
27	MCQ	+2	D
28	MCQ	+1	D
29	NAT	+2	0.9 to 1.1
30	NAT	+2	0.9 to 1.1
31	NAT	+1	0.49 to 0.51
32	MSQ	+2	A, C
33	MCQ	+2	C
34	MCQ	+2	C
35	MCQ	+2	A
36	MCQ	+2	D
37	MCQ	+2	C
38	MCQ	+1	A

Q. No.	Type	Marks	Official Key
39	MCQ	+1	C
40	NAT	+2	1.0 to 1.0
41	NAT	+1	0.5 to 0.5
42	NAT	+1	1.0 to 1.0
43	NAT	+1	0.5 to 0.5
44	MCQ	+2	C
45	MCQ	+2	B
46	MCQ	+1	B
47	MCQ	+1	D
48	NAT	+2	0.10 to 0.11
49	NAT	+2	0.25
50	NAT	+1	1.64 to 1.65
51	MSQ	+2	C;D
52	MCQ	+2	A
53	MCQ	+2	B
54	MCQ	+2	D
55	MCQ	+1	B

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.